4. Which group of the periodic table generally has the lowest electron affinity values?

(A) Group 1 (Alkali metals) (B) Group 2 (Alkaline earth metals)
(C) Group 17 (Halogens) (D) Group 18 (Noble gases)

5. Which of the following is an example of a hard base?

(A) CN^- (B) OH^- (C) I^- (D) SCN^-

Part – B (2 x 10 = 20 Marks)

6. a. i Explain the origin of color in coordination compounds with reference to d-d transitions. Provide an example of a colored complex and explain how d-orbital splitting leads to its observed color. (5 Marks)

ii Calculate the Crystal Field Stabilization Energy (CFSE) in terms of Δ_o and P for the following complexes, assuming an octahedral geometry:

(1) $[Fe(H_2O)_6]^{2+}$ and (2) $[Fe(CN)_6]^{4-}$ (5 Marks)

(OR)

b. Discuss types of isomerism in coordination compounds? Explain structural isomerism in details with suitable examples. (10 Marks)

7 a. i Given the following acids and bases: Ca^{2+}, Cr^{3+}, CO, I^-, Hg^{2+} and OH^-, classify each as hard or soft according to the HSAB principle. Explain your classification. (6 Marks)

ii For a chlorine (Cl) atom, use Slater's rules to calculate the effective nuclear charge for a 3p electron. Show the calculation process. (Z of Cl = 17) (4 Marks)

(OR)

b. Define electron affinity. How does electron affinity change across a period and down a group? Explain the factors influencing electron affinity. (10 Marks)

Q. No.	BL	CO	PO
1	4	1	1
2	2		3
3	3		4
4	2		1
5	2		1

Q. No.	BL	CO	PO
6a (i)	3	1	4
(ii)	2		3
6b	1		4
7a (i)	3		3
(ii)	2		1
7b	1		1

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Set- 2

INTERNAL ASSESSMENT – I [FJ1]

Program	: B. Tech.	Date	: 27-09-2024
Course Code & Title	: 21CYB101J & Chemistry	Duration	: 12.30 – 1.20 PM
Year & Sem	: I Year & I Sem	Max. Marks	: 25 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. What is the magnetic moment of Fe^{3+} in a high-spin octahedral complex? (Z of Fe = 26)

   (A) 1.73 BM (B) 2.83 BM (C) 5.92 BM (D) 0 BM

2. Which of the following is the correct order of ligands in the spectrochemical series from weak field to strong field?

   (A) $I^- < Cl^- < H_2O < NH_3 < CN^-$ (B) $CN^- < NH_3 < H_2O < Cl^- < I^-$

   (C) $Cl^- < I^- < H_2O < NH_3 < CN^-$ (D) $NH_3 < CN^- < H_2O < Cl^- < I^-$

3. Which among the following complexes can exhibit isomerism?

   (A) $[Ag(NH_3)_2]^+$ (B) $[Co(NH_3)_5NO_2]^{2+}$ (C) $[Pt(en)Cl_2]$ (D) $[Co(NH_3)_5Cl]^+$

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4. Which of the following elements has the highest electron affinity?

(A) Oxygen (B) Fluorine (C) Nitrogen (D) Chlorine

5. Which of the following is a soft base?

(A) OH^- (B) F^- (C) I^- (D) NO_3^-

Part – B (2 x 10 = 20 Marks)

6. a. Discuss how CFT explains the splitting of d-orbitals in an octahedral crystal field; calculate CFSE and magnetic moment for high-spin as well as low-spin complexes of metal ion with d^7 configuration. (10 marks)

(OR)

b. i. What is the difference between high-spin and low-spin complexes in terms of color? Use d-d transitions and crystal field theory to explain. (5 Marks)

ii. Provide an example of a coordination compound that exhibits optical isomerism. Explain how the isomers are different and how their optical activity can be observed. (5 Marks)

7 a. i Give HSAB principle. Explain how it categorizes acids and bases and provide an example of a reaction that illustrates the principle. (6 Marks)

ii Calculate the effective nuclear charge experienced by a 3p electron in an argon (Ar) atom using Slater's rules. Show all steps in your calculation. (Z of Ar = 18) (4 marks)

(OR)

b. Define atomic size, explain the factors affecting and discuss the periodic trends in atomic size across a period and down a group in the periodic table. (10 Marks)

Q. No.	BL	CO	PO
1	4	1	1
2	2		1
3	3		3
4	2		4
5	2		4

Q. No.	BL	CO	PO
6a	1	1	1
6b (i)	3		3
(ii)	2		3
7a (i)	1		1
(ii)	2		4
7b	1		1

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Set- 1

RA2411003010217

INTERNAL ASSESSMENT – I [FJI]

Program	: B. Tech.	Date	: 27-09-2024
Course Code & Title	: 21CYB101J & Chemistry	Duration	: 12.30 – 1.20 PM
Year & Sem	: I Year & I Sem	Max. Marks	: 25 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Which of the following ions would have a spin-only magnetic moment of approximately 4.9 BM? (Z of Co = 27, Ni = 28, Mn = 25, Fe = 26)

   (A) Co^{3+} (B) Ni^{2+} (C) Mn^{2+} (D) Fe^{3+}

2. According to the spectrochemical series, which of the following ligands would cause a low-spin configuration in a transition metal complex?

   (A) F^- (B) Cl^- (C) CN^- (D) I^-

3. How many geometrical isomers are possible in a complex of type $[MA_2(L)_2]$, where A is unidentate and L is didentate?

   (A) 0 (B) 2 (C) 3 (D) 4

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Classification of structural and stereoisomerism (types) – 2 marks
Four types of structural isomerism with suitable examples and explanation – 8 marks

7 a. i Given the following acids and bases: : Ca^{2+}, Cr^{3+}, CO, I^-, Hg^{2+} and OH^-, classify each as hard or soft according to the HSAB principle. Explain your classification. (6 Marks)

Ans: Classification – 3 marks, Explanation – 3 marks
Hard Acid: Ca^{2+}, Cr^{3+}
Soft Acid: Hg^{2+}
Hard Base: OH^-
Soft Base: I^-, CO

ii For a chlorine (Cl) atom, use Slater's rules to calculate the effective nuclear charge for a 3p electron. Show the calculation process. (4 Marks)

Ans:
Calculation steps: 3 marks
Z_{eff} value: 1 mark

$\underbrace{1s^2}_{(n-2)}$ $\underbrace{2s^2\ 2p^6}_{(n-1)}$ $\underbrace{3s^2\ 3p^5}_{n}$

Group	no. of electrons	Contribution of each electron to 'S' value	Contribution of a particular group
n	6	0.35	2.1
(n – 1)	8	0.85	6.8
(n – 2)	2	1	2
		S =	10.9

Effective nuclear charge = Z- S = 17 – 10.9

$(Z_{eff})_{Cl}$ = 6.1

(OR)

b. i. Define electron affinity. How does electron affinity change across a period and down a group? Explain the factors influencing electron affinity. (10 Marks)

Ans:
Definition: 2 marks
Trends : 2 marks
Factors: 6 marks

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Set- 2

INTERNAL ASSESSMENT – I [FJ1]-Answer Key

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & 1 Sem

Date: 27-09-2024
Duration:
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

1. (C) 5.92 BM
2. (A) $I^- < Cl^- < H_2O < NH_3 < CN^-$
3. (B) $[Co(NH_3)_5NO_2]^{2+}$
4. (D) Group 18 (Noble gases)
5. (B) OH^-

Part – B (2 x 10 = 20 Marks)

6. a. i Explain the origin of color in coordination compounds with reference to d-d transitions. Provide an example of a colored complex and explain how d-orbital splitting leads to its observed color.

Ans: (5 Marks)

Origin of color (explanation with splitting of d orbitals according to CFT, d-d transition energy corresponds to visible range, complementary color of absorbed light) – 4 marks
Example – 1 mark

ii Calculate the Crystal Field Stabilization Energy (CFSE) in terms of Δ_o and P for the following complexes, assuming an octahedral geometry:

(1) $[Fe(H_2O)_6]^{2+}$ and (2) $[Fe(CN)_6]^{4-}$

Ans: (5 Marks)

(1) – 0.4 Δ_o + 1P ; Calculation with steps – 2.5 marks
(2) – 2.4 Δ_o + 3P ; Calculation with steps – 2.5 marks

(OR)

b. Discuss types of isomerism in coordination compounds? Explain structural isomerism in details with suitable examples.

Ans: (10 Marks)

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Difference in isomers – 2 marks
Observation of optical activity by rotation of polarized light – 1 marks

7 a. i Give HSAB principle. Explain how it categorizes acids and bases and provide an example of a reaction that illustrates the principle. (6 Marks)

Ans:
Principle – 1 marks
Properties of acids and bases – 4 marks (2+2)
One example of reaction – 1 mark

ii Calculate the effective nuclear charge experienced by a 3p electron in an argon (Ar) atom using Slater's rules. Show all steps in your calculation. (4 marks)

Ans: Steps: 3 marks
Zeff value – 1 mark

The electron configuration of argon is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Ar $(1s^2)\ (2s^2 2p^6)\ (3s^2 3p^6)$

7 other 3s 3p electrons = 7 x 0.35 = 2.45

8 2s 2p electrons = 8 x 0.85 =6.80

2 1s electrons = 2 x 1.00 = 2

S = 2.45 + 6.80 + 2 = 11.25

$Z_{eff} = Z - S$
$= 18 - 11.25$
$= 6.75$

(OR)

b. Define atomic size and discuss the periodic trends in atomic size across a period and down a group in the periodic table. Explain the underlying reasons (factors) for these trends (10 Marks)

Ans:
Definition: 2 marks
Trends : 2 marks
Factors: 6 marks

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Set- 1

INTERNAL ASSESSMENT – I [FJ1]- Answer key

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 27-09-2024
Duration:
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

1. (A) Co^{3+}
2. (C) CN^-
3. (B) 2
4. (D) Chlorine
5. (C) I^-

Part – B (2 x 10 = 20 Marks)

6. a. Explain the key postulates of Crystal Field Theory (CFT) for octahedral complexes. Discuss how CFT explains the splitting of d-orbitals in an octahedral crystal field and the formation of high-spin and low-spin complexes. (10 marks)

Ans:
Postulates of CFT – 3 marks;
Explanation of splitting in octahedral complexes with diagram – 4 marks
Formation of high spin and low spin complexes with examples – 3 marks

(OR)

b. i. What is the difference between high-spin and low-spin complexes in terms of color? Use d-d transitions and crystal field theory to explain your answer. (5 Marks)

Ans:
High-Spin Complexes: Small Δ, more unpaired electrons, absorb lower energy light, leading to colors like green or red. **(2.5 marks)**
Low-Spin Complexes: Large Δ, fewer unpaired electrons, absorb higher energy light, leading to colors like yellow or orange. **(2.5 marks)**

ii. Provide an example of a coordination compound that exhibits optical isomerism. Explain how the isomers are different and how their optical activity can be observed. (5 Marks)

Ans:
Any one example with structure – 2 marks

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7. According to the convention, the Daniel cell is represented as:
   (a) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt
   (b) Zn l $ZnSO_4$ll Cu l $CuSO_4$, E = 1.09 volt
   (c) $ZnSO_4$ l Zn ll $CuSO_4$ l Cu, E = 1.09 volt
   (d) Zn l ZnSll $CuSO_4$ l Cu, E = 1.09 volt

8. What type of reaction takes place upon treatment of a ketone with HCN to form a cyanohydrin?
   (a) Nucleophilic addition (b) Nucleophilic substitution
   (c) Electrophilic addition (d) Electrophilic substitution

9. How many aromatic isomers of dibromobenzene exist?
   (a) 2 (b) 3 (c) 4 (d) 6

10. Which of the following is true about Fischer Projection?
   (a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.
   (b) The vertical lines are oriented towards you and the horizontal lines are oriented away from you.
   (c) Both the horizontal and vertical lines are oriented away from you.
   (d) Both the horizontal and vertical lines are oriented towards you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance. (6 Marks)

(ii) For a Daniel Cell involving a cell reaction,

$$Zn_{(s)} + Cu^{2+}{}_{(aq)} \leftrightarrow Zn^{2+}{}_{(aq)} + Cu_{(s)}$$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}{}_{(aq)}$, $Zn^{2+}{}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)

12(a) Write a note on the Pourbaix diagram of iron (Fe). (10 Marks)

(OR)

12(b) (i) Explain the SN^1 mechanism with a suitable example. (6 Marks)

(ii) Draw the Newman and Sawhorse projections for the eclipsed and staggered conformations of ethane. Which conformation is more stable and why? (4 Marks)

No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a (i)	1	2	1
(ii)	2		2
11b (i)	3		3
(ii)	1		1
12a.	2		2
12b (i)	3	3	2
(ii)	2		3

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Set- 1

INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.		
Course Code & Title	: 21CYB101J & Chemistry	Date	: 21-11-2024
Year & Sem.	: I Year & I Sem.	Time	: 8.00-9.00am
		Max. Marks	: 30 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10 Marks)
Answer ALL the questions

1. When equilibrium is reached inside the two half-cells of the electrochemical cell, what is the net voltage across the electrodes?
   (a) > 1 (b) < 1 (c) = 0 (d) Not defined
2. What will be the Gibbs free energy for the reaction of conversion of ATP into ADP at 293 Kelvin, if the change in enthalpy is 19.07 Kcal and the change in entropy is 90 cal per Kelvin.
   (a) 7.3 cal (b) -5.3 cal (c) 7.3 Kcal (d) -7.3 Kcal
3. An oxide layer is considered as protective if ratio of volume of oxide layer to volume of base metal is
   (a) > 1 (b) > 10 (c) < 1 (d) > 3
4. Which of the following factors does not affect the electrode potential of an electrode?
   (a) Nature of the electrode (metal) (b) Temperature of the solution
   (c) Molarity of the solution (d) Size of the electrode
5. Which of the following statement is incorrect?
   (a) At constant pressure, $\Delta H = \Delta E + P\Delta V$
   (b) Thermodynamic symbol for entropy is S.
   (c) Gibbs free energy is a state function.
   (d) For an endothermic process, ΔH is negative.
6. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____ gas.
   (a) Nitrogen (b) Chlorine (c) Oxygen (d) Hydrogen

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6. A process is carried out at constant volume and at constant entropy. It will be spontaneous if:
(a) $\Delta H < 0$ (b) $\Delta U < 0$ (c) $\Delta A < 0$ (d) $\Delta G < 0$

7. The entropy of an isolated system always ____ and reaches ____ when equilibrium is reached.
(a) remains constant, maximum (b) decreases, minimum
(c) increases, maximum (d) decreases, constant

8. The isomers which can be inter converted through rotation around a single bond are termed as:
(a) Tautomers (b) positional isomers (c) conformers (d) diastereomers

9. Which of the following statements is **not** correct?
(a) Fischer projection represents the molecule in an eclipsed conformation
(b) Newman projection can be represented in eclipsed, staggered and skew conformations
(c) Fischer projection of the molecule is its most stable conformation
(d) In Sawhorse projections, the lines are inclined at an angle of 120° to each other

10. The reactivity order of alkyl halides in SN^1 mechanism is
(a) $CH_3X > 1^o > 2^o > 3^o$ (b) $CH_3X < 1^o < 2^o < 3^o$
(c) $CH_3X > 3^o > 2^o > 1^o$ (d) $CH_3X > 3^o > 1^o > 2^o$

Part-B (2 x 10 = 20 Marks)

11(a) (i) Describe the construction and working of an electrochemical cell, provide an example to illustrate its operation. (6 Marks)

(ii) Calculate the entropy change for methane combustion from the given thermodynamic data. The Standard entropies of the substances involved in the above reaction are: ΔS°(J/K. mol)) of $CH_{4(g)}$ =186, $O_{2\,(g)}$ = 205 $CO_{2\,(g)}$ = 214 and $H_2O_{(l)}$ = 70 (4 Marks)

(OR)

11(b)
(i) Derive Nernst equation and explain its significance in electrochemistry. (6 Marks)

(ii) Define entropy and explain its significance in thermodynamic and natural processes. (4Marks)

12(a) Explain mechanisms of dry corrosion and wet corrosion, providing the chemical reaction involved in each process. (10 Marks)

(OR)

12(b)
(i) Explain the addition of HBr to propylene, in the presence and absence of peroxide. (8 Marks)

(ii) Define diastereomers with an example. (2 Marks)

Q. No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a. (i)	1	2	1
(ii)	2		2
11b. (i)	3		3
(ii)	1		1
12a.	2		2
12b. (i)	3	3	3
(ii)	2		4

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Set- 2

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INTERNAL ASSESSMENT – II [FJ-II]

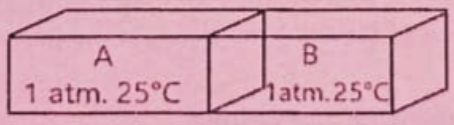
Program	: B. Tech.		
Course Code & Title	: 21CYB101J & Chemistry	Date	: 21-11-2024
Year & Sem.	: I Year & I Sem.	Time	: 8.00-9.00am
		Max. Marks	: 30 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10Marks)
Answer ALL the questions

1. The equilibrium constant for a cell reaction, $Cu_{(g)} + 2Ag^{+}_{(aq)} \leftrightarrow Cu^{2+}_{(aq)} + 2Ag_{(s)}$ is 4×10^{16}. Find $E^{\circ}_{(cell)}$ for the cell reaction.
   (a) 0.63 V (b) 0.49 V (c) 1.23 V (d) 3.24 V
2. An isolated box, equally partitioned, contains two ideal gasses A and B as shown. When the partition is removed, the gases mix. The changes in enthalpy (ΔH) and entropy (ΔS) in the process respectively are:



   (a) Zero, positive (b) Zero, negative (c) Positive, zero (d) Negative, zero
3. Passivity is not reason for inertness of the following:
   (a) Au (b) Al (c) Ti (d) Ni
4. In a reaction, the change in entropy is given as 2.4 cal/K and the change in Gibbs free energy is given as 3.4 kcal, then the change in heat at the temperature of 20 °C is:
   (a) 4.1 cal (b) 3.4 cal (c) 3.4 Kcal (d) 4.1 Kcal
5. Mercury covered by a layer of mercurous chloride in contact with saturated potassium chloride solution is a description of which of the following types of electrodes?
   (a) Chlorine (b) Potassium (c) Calomel (d) Silver/Silver chloride

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12.(a) Explain mechanisms of dry corrosion and wet corrosion, providing the chemical reaction involved in each process. (10 marks)

Ans:

Mechanism of dry corrosion with equation: 2 marks

Three types of wet corrosion with equations: 6 marks

(OR)

12.(b) (i) Explain the addition of HBr to propylene, comparing the reaction in the presence and absence of peroxide. (10 marks)

Ans:

In presence of peroxide (anti-Markovnikov's addition) with example and explanation: 5 marks

In absence of peroxide (Markovniov's addition) with example and explanation: 5 marks

(ii) Define diastereomers with an example.

Ans:

Definition: 1 mark; Example: 1 mark

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Set- 2

ANSWER KEY
INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.		
Course Code & Title	: 21CYB101J & Chemistry	Date	: 21-11-2024
Year & Sem.	: I Year & I Sem.	Time	: 8.00 - 9.00am.
		Max. Marks	: 30 marks

Part-A (10X1 = 10Marks)

1 (b) 0.49 V
2 (a) Zero, positive
3 (a) Au
4 (d) 4.1 Kcal
5 (c) Calomel
6 (b) $\Delta U < 0$
7 (c) increases, maximum
8 (c) conformers

9 (c) Fischer projection of the molecule is its most stable conformation
10 (b) $CH_3X < 1^o < 2^o < 3^o$

Part-B (2 x 10 = 20 Marks)

11. (a) (i) Describe the construction and working of an electrochemical cell, provide an example to illustrate its operation. (6 marks)

Ans:
Neat and labelled diagram:3 marks
Oxidation reduction reactions: 2 marks
Cell notation and emf : 1 mark

(ii) Calculate the entropy change for methane combustion from the given thermodynamic data. The Standard entropies of the substances involved in the above reaction are: ΔS^o(J/K. mol)) of $CH_{4(g)}$ =186, $O_{2\ (g)}$ = 205 $CO_{2\ (g)}$ = 214 and $H_2O_{\ (l)}$ = 70 (4 marks)

Ans:
Balanced chemical equation: 1 marks
Calculation: 3 marks
$\Delta S^o_{reaction}$ = [$\sum$ Standard entropies of products - $\sum$ Standard entropies of reactants]
ΔS^o = [214 + 2* 70] - [186 + 205 * 2] = -242 J/K

(OR)

11.(b) (i) Derive Nernst equation and explain its significance in electrochemistry. (6 marks)

Ans:
Derivation: 6 marks
Applications: 2 marks

(ii) Define entropy and explain its significance in thermodynamic and natural processes. (4 marks)

Ans: Definition: 1 mark; Significance: 3 marks

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12(a) Write a note on the Pourbaix diagram of iron (Fe). (10 Marks)

(i) **Ans: Neat and labelled diagram: 3 Marks**
Explanation of 5 curves with chemical reactions: 3 marks
Explanation of regions 2 marks
Characteristics and significance of diagram : 2 marks

(OR)

12(b) Explain the SN^1 mechanism with a suitable example. (6 Marks)

(i) **Ans: Two step mechanism with suitable example: 4 marks**
Explanation: 2 marks

(ii) Draw the Newman and Sawhorse projections for the eclipsed and staggered conformations of ethane. Which conformation is more stable and why? (4 Marks)

Ans: Newman projection (eclipsed and staggered): 1.5 marks
Sawhorse projection (eclipsed and staggered): 1.5 marks
Explanation of stable conformation with reason: 1 mark

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SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
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Set- 1

ANSWER KEY
INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00 - 9.00am.
Year & Sem.	: I Year & I Sem.	Max. Marks	: 30 marks

Part-A (10X1 = 10 Marks)

1. (c) = 0
2. (d) -7.3 Kcal
3. (a) > 1
4. (d) Size of the electrode
5. (d) For an endothermic process, ΔH is negative.
6. (d) Hydrogen
7. (a) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt
8. (a) Nucleophilic addition
9. (b) 3
10. (a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance.
Ans: Derivation – 5 marks, Significance – 1 Mark

(ii) For a Daniel Cell involving a cell reaction,
$Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

Ans:

$\Delta G^o_{rec} = \Delta G^o_{f(prod)} - \Delta G^o_{f(react)}$

$= (-154 + 0) - (0 + 64.4) = -218.4$ kJ/mol

$E^o = -\Delta G^o_{rec}/nF$

$= 1.13$ V

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

Ans: Explanation with example including working and reference electrode, reactions involved, Cell notation, expression for emf and 3 plots)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)
Ans: Any 4 points – 4 marks

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RA2

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Set-2

INTERNAL ASSESSMENT – I (FJ1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year &Sem: I Year &I Sem

Date: 25-09-2024
Duration: 8.00-8.50am
Max. Marks: 25 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 Among the following base, which one is soft?
(a) NH_3 (b) I^- (c) H_2O (d) OH^-

2 An aqueous solution of Ni^{2+} contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 BM. When ammonia is added in it the magnetic moment of solution_____

(a) will remain the same
(b) will increase from 2.83 BM
(c) will decrease from 2.83 BM
(d) cannot be predicted theoretically.

3 How many geometrical isomers are possible in $[Al(C_2O_4)_3]^{3-}$?
(a) 0 (b) 2 (c) 3 (d) 4

4 $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$ differ in:

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Set- 1

ANSWER KEY
INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00 - 9.00am.
Year & Sem.	: I Year & I Sem.	Max. Marks	: 30 marks

Part-A (10X1 = 10 Marks)

1. (c) = 0
2. (d) -7.3 Kcal
3. (a) > 1
4. (d) Size of the electrode
5. (d) For an endothermic process, ΔH is negative.
6. (d) Hydrogen
7. (a) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt
8. (a) Nucleophilic addition
9. (b) 3
10. (a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance.
Ans: Derivation – 5 marks, Significance – 1 Mark

(ii) For a Daniel Cell involving a cell reaction,
$Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

Ans:

$\Delta G^o_{rec} = \Delta G^o_{f(prod)} - \Delta G^o_{f(react)}$

$= (-154 + 0) - (0 + 64.4) = -218.4$ kJ/mol

$E^o = -\Delta G^o_{rec}/nF$

$= 1.13$ V

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

Ans: Explanation with example including working and reference electrode, reactions involved, Cell notation, expression for emf and 3 plots)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)
Ans: Any 4 points – 4 marks

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2024-25 ODD CSE-Y2

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Set-1

INTERNAL ASSESSMENT–I (FJI)

Program: B.Tech	Date: 25-09-2024
Course Code & Title: 21CYB101J &Chemistry	Duration: 8.00-8.50am
Year&Sem: IYear & ISem	Max.Marks:25marks

Part –A(5x1=5Marks)

Answer ALL The Questions

1 (d) $d_{x^2-y^2}$ and d_{z^2}

2 (b) number of d-electrons

3 (c) $[AlF_6]^{3-}$

4 (a) 2

5 (b)Coordination number

Part– B (2 x10= 20Marks)

6. a.i. Calculate the CFSE of d^4 and d^7 in high spin tetrahedral complexes in terms of Δo

(6 Marks)

Energy level diagram showing electrons distribution over e and t_2 sets for d^4 and d^7 - (2 marks)

CFSE calculation for d^4 ion, electron distribution $-e^2t^2$; CFSE= $-0.18\Delta o$(2Marks)

CFSE calculation for d^7 ion, electron distribution $-e^4t^3$; CFSE= $-0.54\Delta o$(2Marks)

ii. Write short note on linkage and hydrate isomerism in coordination compounds.

(4 Marks)

Explanation on linkage isomerism with examples -2 marks

Explanation on hydrate isomerism with examples -2marks

(OR)

b.i. Calculate the energy of the $[Ti(H_2O)6]^{3+}$ complex with a wave number of 20,000cm^{-1}

(5Marks)

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1 kJ/mol =83.6 cm^{-1} **–1mark**

1 cm^{-1}= 1/83.6kJ/mol **– 1mark**

So, 20,000cm^{-1} =20,000/83.6 kJ/mol=239.23 kJ/mol **–2marks**

Unit **– 1 mark**

ii. Which among the following complexes have large crystal field splitting in each pair with appropriate justification? (5 Marks)

(1)$[Co(H_2O)6]^{2+}$and$[Co(H_2O)6]^{3+}$ (2)$[Co(CN)6]^{3+}$and$[Co(NH3)6]^{3+}$

(1)$[Co(H_2O)6]^{3+}$–1 mark

Explanation based on cation with higher oxidation state has larger value of Δ_o

-1.5marks

(2)$[Co(CN)6]^{3+}$ **-1 mark**

Explanation based on Δ_o increasing with increase of ligand field strength

-1.5 marks

7 a. Give the steps for Slater's rule and using it, calculate Zeff for an electron residing in 2p level of F^-ion (Z for Fluorine atom is 9).

Slater rule steps **-5 marks**

Calculation of Z* for a 2p electron in F^-ion

- **F: $(1s^2)(2s^2,2p^6)$** **-1 mark**
- **Rule2doesnot apply–1Mark**
- **Shielding constant S= (7x0.35) +(2 x0.85)= 4.15** **-2marks**
  Z* =Z-S =9–4.15=4.85 **-1 mark**

(OR)

b.i. What are the characteristics of hard acids? Give few examples -5marks

Characteristics **-2.5 marks**

Examples **-2.5marks**

ii. Calculate the magnetic moment of high spin complexes of Fe^{2+} and Co^{3+} [Z for Fe and Co are 26& 27 respectively]. (5 Marks)

Magnetic moment of Fe^{2+} (d^6 , e^- distribution – $t_{2g}^4 e_g^2$, 4 unpaired electrons,

Magnetic moment: 4.898 BM **-2 marks**

Magnetic moment of Co^{3+}–d^6,e^-distribution– $t_{2g}^4 e_g^2$, 4 unpaired electrons

Magnetic moment: 4.898 BM **-2 marks**

Units: **-1mark**

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Set-2

ANSWER KEY-Internal Assessment – I

Program: B.Tech	Date: 25-09-2024
Course Code & Title:21CYB101J & Chemistry	Duration: 8.00-8.50 am
Year &Sem: I Year & I Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. (b) I^-
2. (a) will remain the same
3. (a) 0
4. (c) Magnetic moment and colour
5. (c) $t_{2g}^3 e_g^1$

Part – B (2 x 10 = 20 Marks)

6. a. i. What are different types of isomerism in coordination compounds? Describe geometrical and optical isomerism with suitable examples. (10 marks)

Ans: Types of isomerism flow chart: 2 marks
Geometrical isomerism in square planar and octahedral complexes, explanation with examples: 4 marks
Optical isomerism explanation with examples: 4 marks

(OR)

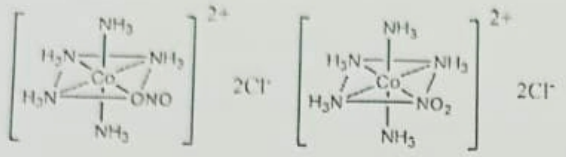
b. i. Explain, why the transition metal coordination compounds with strong field ligands are yellow, orange or red in color, whereas with weak field ligands they are often blue-green, blue or indigo in color. (5 marks)

Ans: In general, **strong-field ligands** cause a large split in the energies of d orbitals of the central metal atom (**large Δ_{oct}**). Transition metal coordination compounds with these ligands **absorb higher-energy violet or blue light and hence are often yellow, orange, or red in colour.** **(2.5 marks)**
On the other hand, coordination compounds of transition metals **with weak-field ligands cause low** split in the energies of d orbitals of the central metal atom (**small Δ_{oct}**). Transition metal coordination compounds with these ligands **absorb lower-energy yellow, orange, or red light and are often blue-green, blue, or indigo** in colour. **(2.5 marks)**

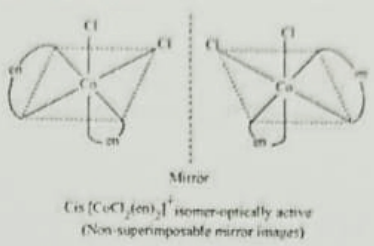
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ii. Draw structures for linkage isomers of $[Co(NH_3)_5NO_2]Cl_2$ and optical isomers of $[CoCl_2(en)_2]^+$. (5 marks)

2.5 Marks



2.5 Marks



Cis $[CoCl_2(en)_2]^+$ isomer-optically active
(Non-superimposable mirror images)

7 a. What are the features of crystal field splitting theory? Calculate CFSE values in terms of Δ_o for high spin and low spin octahedral complexes having d^5 and d^7 configuration. (10 marks)

Ans :Salient features 5 points **-5marks**

Ans:	
d^5 configuration **(2.5 marks)**	d^7 configuration **(2.5 marks)**
Low spin: CFSE = $-2\ \Delta_o + 2P$	**Low spin:** CFSE = $-1.8\ \Delta_o + 3P$
High spin: CFSE = $0\ \Delta_o$	**High spin:** CFSE = $-0.8\ \Delta_o + 2P$

(OR)

b. i. Explain the factors affecting crystal field splitting with suitable examples. (6 marks)

Ans: Any three factors with examples: **each 2 marks**

ii. With examples, define hard base and soft base (4 marks)

Ans :Definition of hard base with 2 examples **- 2marks**

Definition of soft base with 2 examples **- 2marks**

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Set-1

INTERNAL ASSESSMENT – I (FJI)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & 1 Sem
Course Articulation Matrix

Date: 25-09-2024
Duration: 8.00-8.50am
Max. Marks: 25 marks

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 When the valence d orbitals of the central metal ion in octahedral complex are split in energy levels in CFT, which orbitals are raised to higher energy?

(a) d_{xy} and $d_{x^2-y^2}$ (b) d_{xy}, d_{xz} and d_{yz} (c) d_{xz} and d_{yz} (d) $d_{x^2-y^2}$ and d_{z^2}

2 The crystal field splitting energy (Δo) is inversely proportional to

(a) geometry (b) number of d-electrons (c) coordination number (d) oxidation state

3 Which of the following complex is most stable?

(a) $[AlBr_6]^{3-}$ (b) $[AlI_6]^{3-}$ (c) $[AlF_6]^{3-}$ (d) $[AlCl_6]^{3-}$

4 How many geometrical isomers are possible for $[Co(NH_3)_3(NO_2)_3]$ complex?

(a) 2 (b) 3 (c) 4 (d) 0

5 The number of unidentate ligands in the complex ion is called _____

(a) EAN (b) Coordination number (c) Primary valency (d) Oxidation number

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Set-1

INTERNAL ASSESSMENT – I (FJI)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & 1 Sem
Course Articulation Matrix

Date: 25-09-2024
Duration: 8.00-8.50am
Max. Marks: 25 marks

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 When the valence d orbitals of the central metal ion in octahedral complex are split in energy levels in CFT, which orbitals are raised to higher energy?

(a) d_{xy} and $d_{x^2-y^2}$ (b) d_{xy}, d_{xz} and d_{yz} (c) d_{xz} and d_{yz} (d) $d_{x^2-y^2}$ and d_{z^2}

2 The crystal field splitting energy (Δo) is inversely proportional to

(a) geometry (b) number of d-electrons (c) coordination number (d) oxidation state

3 Which of the following complex is most stable?

(a) $[AlBr_6]^{3-}$ (b) $[AlI_6]^{3-}$ (c) $[AlF_6]^{3-}$ (d) $[AlCl_6]^{3-}$

4 How many geometrical isomers are possible for $[Co(NH_3)_3(NO_2)_3]$ complex?

(a) 2 (b) 3 (c) 4 (d) 0

5 The number of unidentate ligands in the complex ion is called _____

(a) EAN (b) Coordination number (c) Primary valency (d) Oxidation number

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Part – B (2 x 10 = 20 Marks)

6. a. i. Calculate the CFSE of d^4 and d^7 in high spin tetrahedral complexes in terms of Δo (6 Marks)

ii. Write short note on linkage and hydrate isomerism in coordination compounds. (4 Marks)

(OR)

b. i. Calculate the energy of the $[Ti(H_2O)_6]^{3+}$ complex with a wave number of 20,000 cm^{-1}. (5 Marks)

ii. Which among the following complexes have large crystal field splitting in each pair with appropriate justification? (5 Marks)

(1) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$ (2) $[Co(CN)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$

7 a. Give the steps for Slater's rule and using it, calculate Z_{eff} for an electron residing in 2p level of F^- ion (Z for Fluorine atom is 9). (10 Marks)

(OR)

b. i. What are the characteristics of hard acids? Give few examples. (5 Marks)

ii. Calculate the magnetic moment of high spin complexes of Fe^{2+} and Co^{3+} [Z for Fe and Co are 26 & 27 respectively]. (5 Marks)

Q.No	BL	CO	PO
1	2	1	4
2	1		1
3	3		3
4	3		3
5	1		1

Q.No	BL	CO	PO
6 a.	3	1	4
	1		1
6b.	3		4
	4		3
7a.	3		3
.			
7b.	2		1
	3		3

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RA2

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Set-2

INTERNAL ASSESSMENT – I (FJ1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year &Sem: 1 Year &I Sem

Date: 25-09-2024
Duration: 8.00-8.50am
Max. Marks: 25 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 Among the following base, which one is soft?
(a) NH_3 (b) I^- (c) H_2O (d) OH^-

2 An aqueous solution of Ni^{2+} contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 BM. When ammonia is added in it the magnetic moment of solution_____

(a) will remain the same
(b) will increase from 2.83 BM
(c) will decrease from 2.83 BM
(d) cannot be predicted theoretically.

3 How many geometrical isomers are possible in $[Al(C_2O_4)_3]^{3-}$?
(a) 0 (b) 2 (c) 3 (d) 4

4 $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$ differ in:

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(a) Geometry and magnetic moment (b) Geometry and hybridization

(c) Magnetic moment and color (d) Hybridization and number of d electrons

5 The electronic configuration of metal atom/ionic octahedral complex with d^4 configuration, if $\Delta_o <$ pairing energy is:

(a) $t_{2g}^4 e_g^0$ (b) $e_g^4 t_{2g}^0$ (c) $t_{2g}^3 e_g^1$ (d) $e_g^2 t_{2g}^2$

Part – B (2 x 10 = 20 Marks)

6. a. i. What are different types of isomerism in coordination compounds? Describe geometrical and optical isomerism with suitable examples. (10 Marks)

(OR)

b. i. Explain, why the transition metal coordination compounds with strong field ligands are yellow, orange or red in color, whereas with weak field ligands they are often blue-green, blue or indigo in color. (5 Marks)

ii. Draw structures for linkage isomers of $[Co(NH_3)_5NO_2]Cl_2$ and optical isomers of $[CoCl_2(en)_2]^+$ (5 Marks)

7 a. What are the features of crystal field splitting theory? Calculate CFSE values in terms of Δ_o for high spin and low spin octahedral complexes having d^5 and d^7 configuration. (10 Marks)

(OR)

b. i. Explain the factors affecting crystal field splitting with suitable examples. (6 Marks)

ii. With examples, define hard base and soft base. (4 Marks)

Q.No	BL	CO	PO
1	1	1	1
2	3		3
3	2		4
4	3		3
5	2		4

Q.No	BL	CO	PO
6 a.	1	1	1
6b.	3		3
	2		4
7a.	3		3
7b.	2		4
	1		1

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Set- 1

ANSWER KEY
INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00 - 9.00am.
Year & Sem.	: I Year & I Sem.	Max. Marks	: 30 marks

Part-A (10X1 = 10 Marks)

1. (c) = 0
2. (d) -7.3 Kcal
3. (a) > 1
4. (d) Size of the electrode
5. (d) For an endothermic process, ΔH is negative.
6. (d) Hydrogen
7. (a) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt
8. (a) Nucleophilic addition
9. (b) 3
10. (a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance.
Ans: Derivation – 5 marks, Significance – 1 Mark

(ii) For a Daniel Cell involving a cell reaction,
$Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

Ans:

$\Delta G^o_{rec} = \Delta G^o_{f(prod)} - \Delta G^o_{f(react)}$

$= (-154 + 0) - (0 + 64.4) = -218.4$ kJ/mol

$E^o = -\Delta G^o_{rec}/nF$

$= 1.13$ V

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

Ans: Explanation with example including working and reference electrode, reactions involved, Cell notation, expression for emf and 3 plots)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)
Ans: Any 4 points – 4 marks

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